

IPs de las PCs

PC	IP	Máscara	Gateway	VLAN
DOC1	172.22.8.10	255.255.255.0	172.22.8.1	20 - Docentes
EST1	172.22.0.10	255.255.248.0	172.22.0.1	10 - Estudiantes
GES1	172.22.10.70	255.255.255.224	172.22.10.65	30 - Gestión
ACA1	172.22.9.10	255.255.255.128	172.22.9.1	70 - Académicos
FIN1	172.22.9.140	255.255.255.128	172.22.9.129	60 - Finanzas
SERVICIOS-DOCKER	172.22.10.10	255.255.255.192	172.22.10.1	80 - Datacenter

ISP

```
enable
configure terminal
hostname ISP
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
interface e0/0
ip address 172.22.10.166 255.255.255.252
description Enlace-hacia-R1Core
no shutdown
interface Loopback0
ip address 8.8.8.8 255.255.255.255
description Simula-Internet
ip route 172.22.0.0 255.254.0.0 172.22.10.165
end
write memory
```

R1-Core

```
enable
configure terminal
hostname R1-Core
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
ipv6 unicast-routing
interface e0/0
no shutdown
ip address 172.22.10.169 255.255.255.252
ipv6 address 2001:25:22:C::1/64
ipv6 enable
duplex full
description Enlace-L3-a-SWM1
interface e0/1
no shutdown
ip address 172.22.10.173 255.255.255.252
ipv6 address 2001:25:22:D::1/64
ipv6 enable
duplex full
description Enlace-L3-a-SWM2
interface e0/2
no shutdown
ip address 172.22.10.165 255.255.255.252
ipv6 address 2001:25:22:B::1/64
ipv6 enable
description Enlace-a-ISP
interface s1/0
no shutdown
ip address 172.22.10.161 255.255.255.252
ipv6 address 2001:25:22:A::1/64
ipv6 enable
clock rate 64000
description Serial-WAN-a-R2
interface Loopback0
ip address 172.22.10.176 255.255.255.255
description Loopback-R1Core
```

```
ip route 0.0.0.0 0.0.0.0 172.22.10.166
router ospf 1
router-id 1.1.1.1
network 172.22.10.160 0.0.0.3 area 0
network 172.22.10.168 0.0.0.3 area 0
network 172.22.10.172 0.0.0.3 area 0
network 172.22.10.176 0.0.0.0 area 0
default-information originate
passive-interface e0/2
end
write memory
```

R2

```
enable
configure terminal
hostname R2
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
ipv6 unicast-routing
interface s1/0
no shutdown
ip address 172.22.10.162 255.255.255.252
ipv6 address 2001:25:22:A::2/64
ipv6 enable
description Serial-WAN-a-R1Core
interface e0/0
no shutdown
description Trunk-a-SWA13
interface e0/0.70
encapsulation dot1Q 70
ip address 172.22.9.1 255.255.255.128
ipv6 address 2001:25:22:3::1/64
description VLAN-70-Academicos
interface e0/0.60
encapsulation dot1Q 60
ip address 172.22.9.129 255.255.255.128
```

```

ipv6 address 2001:25:22:4::1/64
description VLAN-60-Finanzas
interface e0/0.90
encapsulation dot1Q 90
ip address 172.22.10.145 255.255.255.240
ipv6 address 2001:25:22:9::1/64
description VLAN-90-Blackhole-R2
interface e0/1
no shutdown
description Trunk-a-SW-Datacenter
interface e0/1.80
encapsulation dot1Q 80
ip address 172.22.10.1 255.255.255.192
ipv6 address 2001:25:22:5::1/64
description VLAN-80-Datacenter
interface Loopback0
ip address 172.22.10.177 255.255.255.255
description Loopback-R2
router ospf 1
router-id 2.2.2.2
network 172.22.10.160 0.0.0.3 area 0
network 172.22.10.177 0.0.0.0 area 0
network 172.22.9.0 0.0.0.127 area 1
network 172.22.9.128 0.0.0.127 area 1
network 172.22.10.0 0.0.0.63 area 1
network 172.22.10.144 0.0.0.15 area 1
passive-interface e0/0.70
passive-interface e0/0.60
passive-interface e0/0.90
passive-interface e0/1.80
end
write memory

```

SWM1

```

enable
configure terminal
hostname SWM1
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4

```

```
password cisco123
login
transport input telnet ssh
ip routing
ipv6 unicast-routing
vtp domain EMPRESA
vtp mode server
vtp version 2
vtp password cisco123
vlan 10
name Estudiantes
vlan 20
name Docentes
vlan 30
name Gestion
vlan 40
name Voz
vlan 50
name Blackhole_R1
vlan 99
name Native_Mgmt
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50 priority 4096
interface range e0/1-3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,50,99
channel-group 1 mode active
channel-protocol lacp
no shutdown
interface Port-channel1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,50,99
description EtherChannel-LACP-a-SWM2
no shutdown
interface e0/0
no switchport
ip address 172.22.10.170 255.255.255.252
ipv6 address 2001:25:22:C::2/64
ipv6 enable
duplex full
description Enlace-L3-a-R1Core
no shutdown
interface e1/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99
description Trunk-a-SWAC1
no shutdown
interface e1/1
switchport trunk encapsulation dot1q
```

```
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,99
description Trunk-a-SWAC2
no shutdown
interface e1/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 30,40,99
description Trunk-a-SWAC3
no shutdown
interface Vlan10
ip address 172.22.0.2 255.255.248.0
ipv6 address 2001:25:22:1::2/64
ipv6 enable
standby version 2
standby 10 ip 172.22.0.1
standby 10 priority 110
standby 10 preempt
standby 10 timers 3 10
description SVI-VLAN10-Estudiantes
no shutdown
interface Vlan20
ip address 172.22.8.2 255.255.255.0
ipv6 address 2001:25:22:2::2/64
ipv6 enable
standby version 2
standby 20 ip 172.22.8.1
standby 20 priority 110
standby 20 preempt
standby 20 timers 3 10
description SVI-VLAN20-Docentes
no shutdown
interface Vlan30
ip address 172.22.10.66 255.255.255.224
ipv6 address 2001:25:22:6::2/64
ipv6 enable
standby version 2
standby 30 ip 172.22.10.65
standby 30 priority 110
standby 30 preempt
standby 30 timers 3 10
description SVI-VLAN30-Gestion
no shutdown
interface Vlan40
ip address 172.22.10.98 255.255.255.224
ipv6 address 2001:25:22:7::2/64
ipv6 enable
standby version 2
standby 40 ip 172.22.10.97
standby 40 priority 110
standby 40 preempt
standby 40 timers 3 10
description SVI-VLAN40-Voz
```

```
no shutdown
interface Vlan50
ip address 172.22.10.130 255.255.255.240
ipv6 address 2001:25:22:8::2/64
ipv6 enable
standby version 2
standby 50 ip 172.22.10.129
standby 50 priority 110
standby 50 preempt
standby 50 timers 3 10
description SVI-VLAN50-Blackhole-R1
no shutdown
interface Loopback0
ip address 172.22.10.178 255.255.255.255
description Loopback-SWM1
router ospf 1
router-id 3.3.3.3
network 172.22.0.0 0.0.7.255 area 0
network 172.22.8.0 0.0.0.255 area 0
network 172.22.10.64 0.0.0.31 area 0
network 172.22.10.96 0.0.0.31 area 0
network 172.22.10.128 0.0.0.15 area 0
network 172.22.10.168 0.0.0.3 area 0
network 172.22.10.178 0.0.0.0 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan50
end
write memory
```

SWM2

```
enable
configure terminal
hostname SWM2
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
```

```
login
transport input telnet ssh
ip routing
ipv6 unicast-routing
vtp domain EMPRESA
vtp mode server
vtp version 2
vtp password cisco123
vlan 10
name Estudiantes
vlan 20
name Docentes
vlan 30
name Gestion
vlan 40
name Voz
vlan 50
name Blackhole_R1
vlan 99
name Native_Mgmt
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50 priority 8192
interface range e0/1-3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,50,99
channel-group 1 mode active
channel-protocol lacp
no shutdown
interface Port-channel1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,50,99
description EtherChannel-LACP-a-SWM1
no shutdown
interface e0/0
no switchport
ip address 172.22.10.174 255.255.255.252
ipv6 address 2001:25:22:D::2/64
ipv6 enable
duplex full
description Enlace-L3-a-R1Core
no shutdown
interface e1/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99
description Trunk-a-SWAC1
no shutdown
interface e1/1
switchport trunk encapsulation dot1q
switchport mode trunk
```



```
switchport trunk native vlan 99
switchport trunk allowed vlan 10,99
description Trunk-a-SWAC2
no shutdown
interface e1/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 30,40,99
description Trunk-a-SWAC3
no shutdown
interface Vlan10
ip address 172.22.0.3 255.255.248.0
ipv6 address 2001:25:22:1::3/64
ipv6 enable
standby version 2
standby 10 ip 172.22.0.1
standby 10 priority 100
standby 10 preempt
standby 10 timers 3 10
description SVI-VLAN10-Estudiantes
no shutdown
interface Vlan20
ip address 172.22.8.3 255.255.255.0
ipv6 address 2001:25:22:2::3/64
ipv6 enable
standby version 2
standby 20 ip 172.22.8.1
standby 20 priority 100
standby 20 preempt
standby 20 timers 3 10
description SVI-VLAN20-Docentes
no shutdown
interface Vlan30
ip address 172.22.10.67 255.255.255.224
ipv6 address 2001:25:22:6::3/64
ipv6 enable
standby version 2
standby 30 ip 172.22.10.65
standby 30 priority 100
standby 30 preempt
standby 30 timers 3 10
description SVI-VLAN30-Gestion
no shutdown
interface Vlan40
ip address 172.22.10.99 255.255.255.224
ipv6 address 2001:25:22:7::3/64
ipv6 enable
standby version 2
standby 40 ip 172.22.10.97
standby 40 priority 100
standby 40 preempt
standby 40 timers 3 10
description SVI-VLAN40-Voz
no shutdown
```

```
interface Vlan50
ip address 172.22.10.131 255.255.255.240
ipv6 address 2001:25:22:8::3/64
ipv6 enable
standby version 2
standby 50 ip 172.22.10.129
standby 50 priority 100
standby 50 preempt
standby 50 timers 3 10
description SVI-VLAN50-Blackhole-R1
no shutdown
interface Loopback0
ip address 172.22.10.179 255.255.255.255
description Loopback-SWM2
router ospf 1
router-id 4.4.4.4
network 172.22.0.0 0.0.7.255 area 0
network 172.22.8.0 0.0.0.255 area 0
network 172.22.10.64 0.0.0.31 area 0
network 172.22.10.96 0.0.0.31 area 0
network 172.22.10.128 0.0.0.15 area 0
network 172.22.10.172 0.0.0.3 area 0
network 172.22.10.179 0.0.0.0 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan50
end
write memory
```

SWAC1

```
enable
configure terminal
hostname SWAC1
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
```

```

transport input telnet ssh
vtp domain EMPRESA
vtp mode client
vtp version 2
vtp password cisco123
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99
description Trunk-a-SWM1
no shutdown
interface e0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99
description Trunk-a-SWM2
no shutdown
interface e1/0
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-DOC1-VLAN20
no shutdown
end
write memory

```

SWAC2

```

enable
configure terminal
hostname SWAC2
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh

```

```
vtp domain EMPRESA
vtp mode client
vtp version 2
vtp password cisco123
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,99
description Trunk-a-SWM1
no shutdown
interface e0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,99
description Trunk-a-SWM2
no shutdown
interface e2/0
switchport mode access
switchport access vlan 10
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-EST1-VLAN10
no shutdown
end
write memory
```

SWAC3

```
enable
configure terminal
hostname SWAC3
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4          *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
vtp domain EMPRESA
```

```

vtp mode client
vtp version 2
vtp password cisco123
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 30,40,99
description Trunk-a-SWM1
no shutdown
interface e0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 30,40,99
description Trunk-a-SWM2
no shutdown
interface e3/0
switchport mode access
switchport access vlan 30
switchport voice vlan 40
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-GES1-Data30-Voice40
no shutdown
end
write memory

```

SWA13

```

enable
configure terminal
hostname SWA13
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
vtp domain R2BRANCH

```

```

vtp mode transparent
vtp version 2
vlan 60
name Finanzas
vlan 70
name Academicos
vlan 99
name Native_Mgmt
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 60,70,99
description Trunk-a-R2-e0/0
no shutdown
interface e1/0
switchport mode access
switchport access vlan 70
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-ACA1-VLAN70
no shutdown
interface e2/0
switchport mode access
switchport access vlan 60
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-FIN1-VLAN60
no shutdown
end
write memory

```

SW

```

enable
configure terminal
hostname SW
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
* Solo acceso autorizado - EQUIPO 4 *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4

```

```
password cisco123
login
transport input telnet ssh
vtp domain R2BRANCH
vtp mode transparent
vtp version 2
vlan 80
name Datacenter
vlan 99
name Native_Mgmt
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 80,99
description Trunk-a-R2-e0/1
no shutdown
interface e0/1
switchport mode access
switchport access vlan 80
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-SERVICIOS-VLAN80
no shutdown
end
write memory
```

PRUEBAS DE VERIFICACION

PRUEBA 1 — VLANs y VTP

En SWM1:

```
show vlan brief
```

Resultado esperado:

VLAN Name	Status	Ports
1 default	active	
10 Estudiantes	active	
20 Docentes	active	
30 Gestion	active	
40 Voz	active	
50 Blackhole_R1	active	
99 Native_Mgmt	active	

```
show vtp status
```

Resultado esperado:

```
VTP Operating Mode   : Server
VTP Domain Name     : EMPRESA
Number of existing VLANs : 11
Configuration Revision : 7
```

En SWAC1 — verificar que recibio las VLANs:

```
show vtp status
```

Resultado esperado:

```
VTP Operating Mode   : Client
VTP Domain Name     : EMPRESA
Configuration Revision : 7  <- mismo numero que SWM1
```

PRUEBA 2 — EtherChannel

En SWM1 o SWM2:

```
show etherchannel summary
```

Resultado esperado:

```
Group Port-channel Protocol Ports
-----+-----+-----+-----
1   Po1(SU)    LACP   Et0/1(P) Et0/2(P) Et0/3(P)
```

SU = Switch Up. Los tres puertos deben mostrar P (bundled). Si alguno muestra I o D hay problema en ese enlace.

PRUEBA 3 — Spanning Tree

En SWM1:

```
show spanning-tree vlan 10
```

Resultado esperado:

```
VLAN0010
Spanning tree enabled protocol rstp
Root ID  Priority  4106
    This bridge is the root  <- SWM1 ES la raiz
Bridge ID  Priority  4106
```


En SWM2:

```
show spanning-tree vlan 10
```

Resultado esperado:

```
Root ID  Priority  4106
      Cost    47
      Port    65 (Port-channel1) <- llega via EtherChannel
Bridge ID  Priority  8202 <- SWM2 NO es la raiz
```

PRUEBA 4 — HSRP

En SWM1:

```
show standby brief
```

Resultado esperado:

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
VI10	10	110	P	Active	local	172.22.0.3	172.22.0.1
VI20	20	110	P	Active	local	172.22.8.3	172.22.8.1
VI30	30	110	P	Active	local	172.22.10.67	172.22.10.65
VI40	40	110	P	Active	local	172.22.10.99	172.22.10.97
VI50	50	110	P	Active	local	172.22.10.131	172.22.10.129

En SWM2:

```
show standby brief
```

Resultado esperado:

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
VI10	10	100	P	Standby	172.22.0.2	local	172.22.0.1
VI20	20	100	P	Standby	172.22.8.2	local	172.22.8.1
VI30	30	100	P	Standby	172.22.10.66	local	172.22.10.65
VI40	40	100	P	Standby	172.22.10.98	local	172.22.10.97
VI50	50	100	P	Standby	172.22.10.130	local	172.22.10.129

PRUEBA 5 — Failover HSRP

Paso 1 — En DOC1, iniciar ping continuo:

```
ping 172.22.8.1 repeat 9999
```

Paso 2 — En SWM1, apagar la SVI:

```
enable
configure terminal
interface Vlan20
shutdown
```

Paso 3 — En SWM2 verificar que tomo el rol Active:

```
show standby brief
```

Resultado esperado:

```
VI20 20 100 P Active local unknown 172.22.8.1
```

Paso 4 — El ping debe recuperarse en menos de 10 segundos:

```
!!!!!!!!!!!!.....!!!!!!!!!!!!
```

^

maximo 3 puntos
de perdida = correcto

Paso 5 — Restaurar SWM1:

```
interface Vlan20
no shutdown
```

SWM1 recupera automaticamente el rol Active por el preempt.

PRUEBA 6 — OSPF Vecinos

En R1-Core:

```
show ip ospf neighbor
```

Resultado esperado:

Neighbor ID	State	Address	Interface
3.3.3.3	FULL	172.22.10.170	Ethernet0/0
4.4.4.4	FULL	172.22.10.174	Ethernet0/1
2.2.2.2	FULL	172.22.10.162	Serial1/0

Todos deben mostrar FULL. Si aparece INIT o EXSTART hay un problema.

PRUEBA 7 — Ruta Predeterminada OSPF

En SWM1:

show ip route 0.0.0.0

Resultado esperado:

O*E2 0.0.0.0/0 [110/1] via 172.22.10.169, Ethernet0/0

En R2:

show ip route 0.0.0.0

Resultado esperado:

O*E2 0.0.0.0/0 [110/1] via 172.22.10.161, Serial1/0

PRUEBA 8 — Conectividad End-to-End

Desde DOC1 (172.22.8.10) al gateway HSRP:

ping 172.22.8.1

Resultado esperado:

!!!! <- 5 signos = 100% exito

Desde DOC1 a EST1 (inter-VLAN):

ping 172.22.0.10

Resultado esperado:

!!!!

Desde DOC1 a ACA1 (campus a sucursal via OSPF):

ping 172.22.9.10

Resultado esperado:

!!!!

Traceroute desde DOC1 a ACA1:

traceroute 172.22.9.10

Resultado esperado:

```
1 172.22.8.1 1 msec <- HSRP VIP SWM1/SWM2
2 172.22.10.169 2 msec <- R1-Core
3 172.22.10.162 3 msec <- R2
4 172.22.9.10 4 msec <- ACA1
```

Ping a Internet:

ping 8.8.8.8

Resultado esperado:

```
!!!!
Success rate is 100 percent (5/5)
```

Tabla Resumen de Pruebas

Prueba	Comando	Señal OK	Señal de fallo
VTP	show vtp status	Mismo revision en clients	Revision 0 en clients
EtherChannel	show etherchannel summary	Po1 SU, puertos P	Puertos I o D
STP	show spanning-tree vlan 10	SWM1 = root	SWM2 aparece como root
HSRP	show standby brief	SWM1 Active, SWM2 Standby en 5 grupos	Estado Init o Listen
Failover HSRP	show standby brief en SWM2	SWM2 pasa a Active en menos 10s	No cambia estado
OSPF	show ip ospf neighbor	Todos en FULL	INIT o EXSTART
Ruta default Internet	show ip route 0.0.0.0 ping 8.8.8.8	O*E2 en SWM1 y R2 !!!! 100%	No aparece la ruta Request timeout